CLLmap — foundation-model performance

Main classifiers — Accuracy

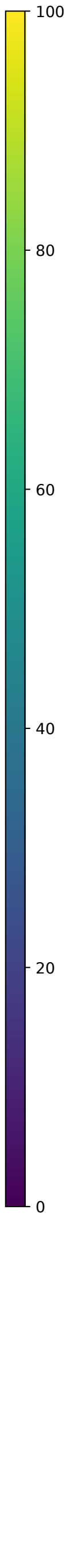
	Omiclass	Logistic Regression
DE baseline		
DE	95.2 ±2.4	N/A
DE · corrected	95.8 ±2.4	N/A
BulkFormer		
Full	66.7	73.6
Full · corrected	82.8	79.3
PCA	72.4	57.5
PCA · corrected	78.2	80.5
Supervised	69.0	79.3
Supervised · corrected	81.6	80.5
Supervised + PCA	72.4	79.3
Supervised + PCA · corrected	75.9	82.8
CellFM 1-view		
Full	52.9	60.9
Full · corrected	52.9	60.9
PCA	58.6	59.8
PCA · corrected	50.6	59.8
Supervised	49.4	50.6
Supervised · corrected	50.6	50.6
Supervised + PCA	51.7	52.9
Supervised + PCA · corrected	55.2	51.7
CellFM 20-view		
Full	74.7	81.6
Full · corrected	82.8	81.6
PCA	82.8	86.2
PCA · corrected	81.6	86.2
Supervised	74.7	80.5
Supervised · corrected	67.8	77.0
Supervised + PCA	66.7	77.0
Supervised + PCA · corrected	67.8	78.2
BulkRNABert		
Full	65.5	79.3
Full · corrected	79.3	82.8
PCA	65.5	70.1
PCA · corrected	77.0	69.0
Supervised	72.4	72.4
Supervised · corrected	78.2	81.6
Supervised + PCA	64.4	75.9
Supervised + PCA · corrected	75.9	75.9

Main classifiers — Macro-F1

	Omiclass	Logistic Regression
DE baseline		
DE	95.2 ±2.4	N/A
DE · corrected	95.8 ±2.4	N/A
BulkFormer		
Full	66.6	73.5
Full · corrected	82.2	78.5
PCA	72.2	56.6
PCA · corrected	77.7	79.8
Supervised	68.8	79.0
Supervised · corrected	81.0	80.1
Supervised + PCA	72.4	78.5
Supervised + PCA · corrected	75.4	82.0
CellFM 1-view		
Full	50.2	56.0
Full · corrected	45.5	56.0
PCA	53.4	55.1
PCA · corrected	41.7	55.1
Supervised	40.9	43.9
Supervised · corrected	43.9	43.9
Supervised + PCA	44.7	46.5
Supervised + PCA · corrected	43.2	46.6
CellFM 20-view		
Full	74.3	81.2
Full · corrected	82.2	81.2
PCA	82.4	85.9
PCA · corrected	81.3	85.9
Supervised	74.4	79.8
Supervised · corrected	67.6	76.5
Supervised + PCA	66.6	76.6
Supervised + PCA · corrected	67.8	77.9
BulkRNABert		
Full	65.5	79.1
Full · corrected	77.4	81.6
PCA	65.5	70.1
PCA · corrected	75.3	68.9
Supervised	72.3	72.3
Supervised · corrected	75.6	80.2
Supervised + PCA	64.3	75.8
Supervised + PCA · corrected	74.8	75.4

Additional classifiers — Accuracy

	MLP	MLP + SMOTE	SVM
BulkFormer			
Full	79.3	44.8	81.6
Full · corrected	75.9	59.8	77.0
PCA	67.8	64.4	67.8
PCA · corrected	74.7	75.9	80.5
Supervised	77.0	74.7	74.7
Supervised · corrected	81.6	71.3	79.3
Supervised + PCA	78.2	78.2	73.6
Supervised + PCA · corrected	80.5	83.9	80.5
Full tokens	83.9	N/A	N/A
Full tokens · corrected	87.4	N/A	N/A
CellFM 1-view			
Full	55.2	66.7	56.3
Full · corrected	62.1	57.5	56.3
PCA	49.4	51.7	62.1
PCA · corrected	50.6	48.3	62.1
Supervised	54.0	54.0	49.4
Supervised · corrected	52.9	57.5	49.4
Supervised + PCA	51.7	54.0	54.0
Supervised + PCA · corrected	47.1	44.8	52.9
Full tokens	54.0	N/A	N/A
Full tokens · corrected	52.9	N/A	N/A
CellFM 20-view			
Full	77.0	66.7	83.9
Full · corrected	78.2	78.2	83.9
PCA	87.4	86.2	88.5
PCA · corrected	85.1	86.2	88.5
Supervised	79.3	71.3	77.0
Supervised · corrected	80.5	69.0	79.3
Supervised + PCA	75.9	73.6	73.6
Supervised + PCA · corrected	75.9	72.4	77.0
Full tokens	93.1	N/A	N/A
Full tokens · corrected	90.8	N/A	N/A
BulkRNABert			
Full	64.4	59.8	69.0
Full · corrected	86.2	80.5	77.0
PCA	78.2	57.5	74.7
PCA · corrected	77.0	78.2	69.0
Supervised	54.0	59.8	65.5
Supervised · corrected	72.4	85.1	75.9
Supervised + PCA	69.0	69.0	69.0
Supervised + PCA · corrected	81.6	73.6	67.8

Additional classifiers — Macro-F1

	MLP	MLP + SMOTE	SVM
BulkFormer			
Full	79.0	39.9	81.3
Full · corrected	74.5	58.4	76.3
PCA	67.8	64.3	67.8
PCA · corrected	73.9	74.8	80.1
Supervised	74.6	74.4	73.4
Supervised · corrected	79.6	71.2	78.5
Supervised + PCA	77.2	77.4	73.1
Supervised + PCA · corrected	80.1	83.4	80.4
Full tokens	83.4	N/A	N/A
Full tokens · corrected	86.8	N/A	N/A
CellFM 1-view			
Full	55.1	63.4	47.9
Full · corrected	59.5	56.0	47.9
PCA	46.9	49.3	51.9
PCA · corrected	47.8	46.5	51.9
Supervised	51.2	51.7	42.1
Supervised · corrected	49.0	54.6	42.1
Supervised + PCA	48.1	49.9	49.9
Supervised + PCA · corrected	41.5	38.9	49.0
Full tokens	53.9	N/A	N/A
Full tokens · corrected	52.8	N/A	N/A
CellFM 20-view			
Full	77.0	66.5	83.4
Full · corrected	78.1	78.1	83.4
PCA	86.9	85.7	88.0
PCA · corrected	84.5	85.7	88.0
Supervised	78.7	71.2	76.5
Supervised · corrected	80.1	68.9	78.7
Supervised + PCA	75.7	73.5	72.9
Supervised + PCA · corrected	75.4	72.1	76.1
Full tokens	92.7	N/A	N/A
Full tokens · corrected	90.0	N/A	N/A
BulkRNABert			
Full	64.3	59.2	68.8
Full · corrected	85.8	80.2	74.9
PCA	78.0	56.3	74.7
PCA · corrected	76.9	78.1	68.8
Supervised	52.2	59.2	65.4
Supervised · corrected	72.4	84.7	74.2
Supervised + PCA	69.0	68.9	68.9
Supervised + PCA · corrected	81.3	73.5	67.6